Locality Dominance in Hyperspectral and Multimodal Land-Cover Classification

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***Abstract*—Hyperspectral and multimodal land-cover classification methods typically rely on high-dimensional per-pixel spectral representations and increasingly complex fusion strategies. In this work, we demonstrate that a compact neighborhood-statistic representation—computed entirely from local spatial context—outperforms substantially higher-dimensional spectral baselines across three benchmarks. Using stratified 10% training splits, a 10–14 dimensional patch-statistics feature vector (mean, standard deviation, gradient magnitude, and PCA-derived neighborhood descriptors) achieves +3.0–11.1 percentage point (pp) improvements in overall accuracy (OA) and +6.3–16.1pp in average accuracy (AA) relative to 30–200 dimensional spectral baselines on Houston (multimodal), Indian Pines, and PaviaU datasets. Notably, gains in AA exceed gains in OA, indicating improved minority-class performance. Accuracy saturates at a small, scene-dependent neighborhood scale (5×5–7×7), beyond which larger windows offer no consistent benefit. In contrast, uncertainty-driven gating and spectral transform augmentations provide negligible or negative impact once spatial context is encoded. These results suggest that local second-order spatial statistics constitute a sufficient and dominant representation for class separation in standard hyperspectral benchmarks, substantially reducing feature dimensionality while improving performance.**

Introduction

Hyperspectral image (HSI) and multimodal remote sensing classification remain central tasks in land-cover analysis. Traditional approaches emphasize high-dimensional spectral representations, often employing dimensionality reduction (e.g., PCA), spectral transforms, or increasingly complex multimodal fusion strategies. More recently, deep convolutional architectures have demonstrated strong performance by implicitly exploiting spatial context through learned receptive fields. Despite these advances, the relative importance of spectral transforms versus explicit spatial aggregation remains underexplored.

Most per-pixel spectral classifiers operate on high-dimensional feature vectors (100–200 bands), implicitly assuming that class separability is encoded primarily in spectral signatures. However, real-world scenes exhibit substantial intra-class spectral variability arising from illumination changes, mixed pixels, and local texture effects. Consequently, pixel-wise models frequently require either high model capacity or complex fusion mechanisms to stabilize predictions.

In this work, we revisit a fundamental question: **Is high-dimensional spectral encoding necessary once local spatial context is explicitly represented?**

We evaluate a compact neighborhood-statistics representation constructed entirely from local spatial summary features—mean, standard deviation, gradient magnitude, and PCA-derived descriptors computed within small spatial windows. The resulting feature vector is 10–14 dimensions, independent of the number of spectral bands.

Across three benchmarks—Houston (multimodal HSI+MS+LiDAR), Indian Pines, and PaviaU—using stratified 10% training splits, we observe that this neighborhood-based representation consistently outperforms 30–200 dimensional spectral baselines. Improvements range from +3.0 to +11.1pp in overall accuracy and +6.3 to +16.1pp in average accuracy, with substantial gains in minority-class performance. Accuracy saturates at a small, scene-dependent window size (5×5–7×7), indicating the existence of a minimal sufficient spatial scale.

We further show that uncertainty-driven gating strategies do not improve performance in the multimodal setting and that spectral transform augmentations become redundant once spatial context is encoded.

The primary contributions of this work are:

1. **Neighborhood Sufficiency Principle (empirical):** A compact local-statistics representation achieves performance comparable to or exceeding high-dimensional spectral encodings across datasets.
2. **Minimal Spatial Scale Identification:** Classification accuracy saturates at a small, scene-dependent neighborhood radius r*, ranging from 5×5 to 7×7 across tested benchmarks.
3. **Minority-Class Stabilization:** Neighborhood statistics disproportionately improve average accuracy (ΔAA > ΔOA), mitigating majority-class bias inherent in high-dimensional spectral models.
4. **Negative Fusion Result:** Uncertainty-based gating (entropy, margin, Gini) does not confer benefit in the evaluated multimodal benchmark; gate confidence exhibits slight negative correlation with classification correctness.

These findings suggest that local second-order spatial statistics constitute a dominant representational mechanism for standard hyperspectral benchmarks, with significant dimensionality and computational advantages.

Related Work

Pixel-wise Spectral Classification

Classical hyperspectral image (HSI) classification methods operate on per-pixel spectral signatures, often comprising 100–200 bands. Dimensionality reduction techniques such as principal component analysis (PCA), linear discriminant analysis (LDA), and band selection methods are commonly applied prior to classification using support vector machines (SVM), random forests (RF), or k-nearest neighbors. While effective in controlled settings, pixel-wise approaches are sensitive to intra-class spectral variability and mixed pixels, often requiring high model capacity to compensate.

Our spectral baselines (PCA-30 and raw spectral RF) follow this standard protocol to provide a direct comparison against neighborhood-based representations.

Spectral–Spatial Feature Extraction

To address the limitations of pixel-only models, numerous works incorporate spatial context through handcrafted texture descriptors, morphological profiles, and spatial filtering techniques. Extended morphological profiles (EMP), gray-level co-occurrence matrix (GLCM) features, and local binary patterns (LBP) have demonstrated that second-order spatial statistics improve classification stability.

More recently, patch-based feature extraction and joint spectral–spatial descriptors have become standard, particularly in conjunction with deep architectures. However, many methods either dramatically increase feature dimensionality or rely on learned convolutional filters.

In contrast, the present work isolates a minimal set of low-dimensional neighborhood summary statistics and evaluates their sufficiency without deep models or high-capacity transforms.

Convolutional Neural Networks for HSI

Convolutional neural networks (CNNs) have achieved state-of-the-art performance in hyperspectral classification by learning hierarchical spectral–spatial filters. These architectures implicitly exploit local receptive fields, progressively aggregating spatial information across layers.

While CNNs demonstrate strong performance, they introduce significant computational overhead and model complexity. The present study investigates whether the performance gains attributed to deep models may arise primarily from local spatial aggregation rather than from deep spectral transformation capacity.

Our results suggest that small fixed neighborhoods (5×5–7×7) with simple second-order statistics capture much of the discriminative power typically associated with convolutional architectures on standard benchmarks.

Multimodal Fusion and Gating

Multimodal remote sensing benchmarks (e.g., HSI + LiDAR + multispectral) have motivated diverse fusion strategies, including feature concatenation, late fusion, attention mechanisms, and uncertainty-driven gating. Entropy-based or margin-based gating schemes aim to weight modalities according to classifier confidence.

However, uncertainty estimates may not align with true class-separating invariants in complex scenes. Our results provide empirical evidence that uncertainty-driven gating can degrade performance and that amplitude-based heuristics provide only modest improvement relative to explicit spatial encoding.

Methods

Datasets

We evaluate on three benchmarks:

- **Houston Multimodal:** Hyperspectral (144 bands), multispectral (8 bands), and LiDAR elevation data with 15 classes. Pre-extracted 11×11 patches are provided; 2832 labeled samples total.
- **Indian Pines:** 145×145 HSI with 200 spectral bands and 16 classes (10249 labeled pixels).
- **PaviaU:** 610×340 HSI with 103 spectral bands and 9 classes (42776 labeled pixels).

For Indian Pines and PaviaU, patches are extracted dynamically from the full image cube using reflect padding at image borders.

Train–Test Protocol

For Indian Pines and PaviaU, we use stratified random splits with 10% of labeled pixels per class allocated to training and 90% to testing, following standard literature protocol. For Houston multimodal, we use a 70%/30% stratified split. All reported results use identical splits across methods for fair comparison. For Houston and Indian Pines, experiments are repeated across three random seeds (42, 1, 7) to assess stability.

PCA transformations are fitted exclusively on the training set and applied to test data to prevent leakage. Patch-level PCA statistics are likewise fitted on training patch pixels only.

Spectral Baselines

Two spectral baselines are evaluated:

1. **PCA-k spectral baseline:** PCA reduced to k components (k=30 for HSI-only datasets, k=50 for Houston), classified using Random Forest (200 estimators).
2. **Raw spectral baseline:** All bands (103–200) used directly with Random Forest.

For Houston multimodal, a 59-dimensional concatenation baseline (HSI PCA-50 + LiDAR + MS) is used as the

primary multimodal reference, along with area-weighted gating variants.

Neighborhood-Statistic Representation

For each labeled pixel, a square window of size r×r centered at the pixel is extracted. The following features are computed over the neighborhood:

- Mean over all spectral bands (flattened patch)
- Standard deviation over all spectral bands
- Gradient magnitude (finite differences on mean-pooled spectral intensity at patch center)
- Mean and standard deviation of the first three PCA components computed over the r×r patch pixels (6 features)

For the multimodal Houston dataset, additional modality-specific statistics are included: LiDAR center-pixel residual (center minus patch mean), MS gradient magnitude, and per-modality mean/std. This yields a 14-dimensional feature vector for Houston and a 10-dimensional vector for HSI-only datasets.

Window sizes r ∈ {3, 5, 7, 11} are evaluated to identify the minimal sufficient spatial scale. The chosen center for all crops is pixel position (5, 5) within the 11×11 pre-extracted patch for Houston, and the labeled pixel center for dynamically extracted patches.

Uncertainty Gating (Houston Only)

For the Houston multimodal dataset, per-modality Random Forest classifiers are trained and used to compute uncertainty signals per sample: entropy (H = -$\sum$ p log p), margin (top-1 minus top-2 probability), and Gini impurity (1 - $\sum p^2$). Gate weights are computed as softmax(u/T) where u is the uncertainty signal and T is a temperature parameter. Sweeps over T ∈ {0.1, 0.5, 1.0, 2.0, 5.0} are conducted.

Gate diagnostics reported include: per-modality mean gate weight, saturation rate (fraction of samples where any gate weight > 0.9 or < 0.1), and Pearson correlation between maximum gate weight and per-sample classification correctness.

Classification Model

Random Forest classifiers (200 estimators, fixed random seed) are used for all primary comparisons to isolate representational effects from model capacity. Experiments with gradient-boosted decision trees (HistGradientBoostingClassifier) confirm that the ranking between representations is stable across classifier choices (difference $\leq$ 0.24pp).

Evaluation Metrics

Performance is measured using:

- **Overall Accuracy (OA):** Fraction of correctly classified test pixels.
- **Average Accuracy (AA):** Mean of per-class accuracies; emphasizes minority-class performance under class imbalance.
- **Cohen's kappa coefficient (κ):** Agreement corrected for chance.

Average Accuracy is emphasized throughout to assess whether gains are broad-based or driven by dominant classes.

Results

Spectral Baselines and Gating Methods (Houston Multimodal)

Uncertainty-driven gating degrades classification performance on the Houston multimodal benchmark. Entropy gating (T = 1.0) reduces overall accuracy (OA) by 9.11 percentage points (pp) relative to the 59-dimensional concatenation baseline (OA = 96.34%), and margin gating at T = 5.0 yields a further -0.48pp. The failure mechanism is structural: the correlation between gate confidence (maximum modality weight) and per-sample classification correctness is corr(max_w, correct) ≈ -0.08, indicating that uncertainty is not a valid fusion invariant on this manifold. The gate fires hardest precisely where the classifier is most likely to be wrong, producing category-level damage concentrated in spectrally ambiguous classes 10–13, which suffer declines of 14–31pp relative to the unweighted baseline. Area gating, which uses amplitude as a proxy for signal stability rather than classifier uncertainty, is the sole gating variant to improve over concat, providing a modest +0.35pp gain (OA = 96.69%). These results establish that naive uncertainty quantification does not confer fusion benefit in this multimodal scene configuration.

Neighborhood-Statistic Representation

Rather than aggregating modality-specific spectral transforms, the patch-statistics representation encodes local spatial context directly from the hyperspectral cube. Each sample is described by a 10–14-dimensional feature vector comprising patch mean, standard deviation, gradient magnitude, and the leading PCA components computed within a square spatial window centered on the labeled pixel. This representation is constructed without any center-pixel spectral information; the entire feature budget is allocated to neighborhood summary statistics.

On the Houston multimodal benchmark, the patch-only configuration (14D, 5×5 window) achieves a mean OA of 99.37% across three independent random seeds (seed = 42, 1, 7), compared to 96.34% for the 59-dimensional concatenation baseline, a gain of +3.03pp (+2.7% relative improvement). Full per-seed and per-method results are reported in Table 1. Augmenting the patch feature vector with chromogeometry descriptors and amplitude anchors (27D total) reduces mean OA by 0.47pp relative to patch-only, confirming that DFT-derived quadrances and spectral centroids are redundant once local spatial statistics are present. Substituting a gradient-boosted

decision tree (GBDT) classifier for the random forest adds at most 0.24pp and does not change the ranking of feature configurations. The full patch-only pipeline requires 0.45 seconds of training time on this dataset, reflecting the low dimensionality of the representation.

Optimal Neighborhood Scale

A patch-size sweep on the Houston scene identifies r* = 5×5 as the optimal window. At 3×3, patch-only OA is 97.64%, a useful improvement over the 96.34% concat baseline but 2.24pp below the 5×5 result, indicating that a three-pixel window is insufficient to capture the spatial homogeneity structure of this scene. Performance at 5×5 (OA = 99.88%, seed = 42) and 11×11 (OA = 99.17%) are within the range of seed-to-seed variance, placing the plateau at r ≥ 5 on this benchmark. The absence of further gain beyond 5×5 is consistent with class-boundary contamination: as the window expands, mixed-class neighborhoods introduce spectral mixture bias that offsets the variance reduction benefit of averaging over a larger sample of within-class pixels.

Cross-Dataset Generalization (Indian Pines and PaviaU)

The generalization experiment, conducted on Indian Pines (145×145 pixels, 200 bands, 16 classes, 10% training) and PaviaU (610×340 pixels, 103 bands, 9 classes, 10% training), constitutes the primary evidence for locality dominance as a scene-independent phenomenon.

On Indian Pines, a 30-dimensional PCA spectral representation applied to the center pixel achieves OA = 70.66%, AA = 56.39%, $\kappa = 0.657$ as the reference spectral baseline. The 10-dimensional patch-only representation at 7×7 achieves OA = 81.00%, AA = 72.46%, $\kappa = 0.782$ (seed = 42), with a mean OA of 81.05% across seeds 42, 1, and 7, confirming seed stability. The gain in average accuracy (ΔAA = +16.08pp) substantially exceeds the gain in overall accuracy (ΔOA = +10.34pp), establishing that improvement is broad-based across classes and disproportionately benefits rare and spectrally ambiguous categories. This asymmetry is diagnostic: the 200-dimensional raw spectral baseline achieves OA = 75.02% but only AA = 60.18%, reflecting majority-class dominance that the spatial representation resolves. On PaviaU, the same patch-only configuration (10D, 7×7) yields OA = 91.98%, AA = 87.96%, $\kappa = 0.893$, compared to the PCA-30 baseline of OA = 88.07%, AA = 81.61%, $\kappa = 0.837$, a gain of ΔOA = +3.91pp and ΔAA = +6.35pp. Full OA, AA, and κ results for both datasets across all patch sizes are reported in Table 2 and Table 3.

The optimal window on both hyperspectral-only benchmarks is r* = 7×7, one step larger than the r* = 5×5 optimum on the Houston multimodal scene. This shift is consistent with the larger spatial homogeneity scale expected in agricultural landscapes (Indian Pines) and structured urban blocks (PaviaU) relative to the densely textured Houston scene, where the multimodal LiDAR channel provides additional structural disambiguation.

Summary

Across all three datasets and a total of nine seed-configuration combinations, a 10–14-dimensional neighborhood-statistic representation consistently and substantially outperforms spectral baselines ranging from 30 to 200 dimensions, achieving a feature-dimension reduction of 76–95% while improving OA by 3.0–11.1pp and AA by 6.3–16.1pp. The neighborhood operator is the dominant contributor to classification performance; spectral transforms, uncertainty-based modality gating, and chromogeometry descriptors are each redundant or mildly harmful once spatial context is encoded. The optimal neighborhood scale r* is scene-dependent and tracks the spatial homogeneity scale of the land-cover pattern, saturating at 5×5 for dense multimodal urban imagery and at 7×7 for hyperspectral agricultural and urban-block scenes.

Discussion

Why Local Neighborhood Statistics Outperform Spectral Transforms

The central empirical finding across all three datasets is that low-dimensional neighborhood statistics outperform substantially higher-dimensional per-pixel spectral representations. This behavior can be understood through variance reduction and spatial stationarity.

Let a pixel spectrum be modeled as:

$$x_i = \mu_c + \varepsilon_i$$

where μ_c is the class-conditional mean and ε_i captures illumination variability, sensor noise, subpixel mixing, and fine-scale texture. Pixel-wise classifiers must separate classes under high within-class variance.

By contrast, the neighborhood operator computes summary statistics over an r×r window:

$$\bar{x}_i = \frac{1}{|N_r|} \sum_{j \in N_r(i)} x_j$$

Under spatial coherence assumptions, variance shrinks approximately as 1/|N_r|, while the class mean remains stable. The representation therefore increases signal-to-noise ratio without increasing feature dimensionality.

The PCA-derived patch statistics and local variance features additionally encode second-order texture structure, capturing discriminative information that spectral basis transforms (e.g., DFT-derived quadrances) do not uniquely provide once spatial context is incorporated.

The Neighborhood Sufficiency Principle (Empirical)

Across datasets, performance improves monotonically with window size up to a scene-dependent threshold r*, beyond which accuracy plateaus or slightly regresses.

This behavior is consistent with two competing effects:

1. **Variance reduction** within homogeneous class regions.
2. **Boundary contamination** when windows span multiple classes.

We therefore observe a minimal sufficient neighborhood scale:

$$r^* \approx \text{class spatial coherence radius}$$

Empirically: - Houston multimodal: r* ≈ 5×5 - Indian Pines: r* ≈ 7×7 - PaviaU: r* ≈ 7×7

These differences align with known scene structure: agricultural fields and urban blocks exhibit larger spatial homogeneity than dense mixed scenes.

Neighborhood Sufficiency Principle (empirical):

> A compact local-statistics representation computed over a small, scene-dependent neighborhood achieves performance comparable to or exceeding high-dimensional per-pixel spectral representations, with performance saturating beyond a bounded spatial scale r*.

Minority-Class Rescue and Class Imbalance

On Indian Pines, the improvement in average accuracy (AA) substantially exceeds the improvement in overall accuracy (OA). This indicates that gains are not limited to dominant classes but instead reflect broad improvements across rare and spectrally ambiguous categories.

Pixel-only spectral models exhibit majority-class bias: OA increases while AA remains suppressed. Neighborhood statistics mitigate this effect by stabilizing minority-class signatures through local averaging and texture encoding.

This suggests that spatial context functions not only as a noise-reduction operator but also as a class-balance equalizer.

Why Uncertainty Gating Fails

Uncertainty-based gating assumes that per-modality classifier confidence is aligned with classification correctness. However, in the Houston multimodal dataset, gate confidence exhibits a slight negative correlation with correctness (corr ≈ -0.08). This indicates that ambiguity is intrinsic to the class manifold rather than a reliable fusion signal.

Area-based gating succeeds modestly because amplitude proxies local signal stability. However, even this approach is dominated by explicit spatial encoding.

Thus, uncertainty is not a reliable invariant for multimodal fusion in these scenes; spatial locality is.

Relation to Deep Learning and Texture Theory

The empirical dominance of small neighborhood statistics aligns with classical texture descriptors (e.g., second-order statistics) and with convolutional neural network receptive field theory.

CNNs implicitly learn spatial filters that capture local mean, variance, and gradient information over small receptive fields. The present results show that much of this benefit can be achieved explicitly using handcrafted low-dimensional statistics, without deep architectures.

This suggests that the primary structural advantage of CNN-based hyperspectral models may lie in their implicit spatial aggregation rather than in high-capacity spectral transforms.

Limitations and Scope

The findings are empirical and evaluated on three benchmarks. While locality dominance is consistent across datasets tested, the optimal window size varies with scene structure. Further evaluation on additional multimodal benchmarks and scenes with highly fragmented spatial patterns would strengthen generality claims.

Additionally, the current work uses fixed window shapes and summary statistics; adaptive or anisotropic neighborhoods may further improve performance.

Implications

These results suggest a design principle for hyperspectral and multimodal classification:

1. Encode spatial context first.
2. Keep representation compact.
3. Introduce spectral transforms only if they add orthogonal information beyond local statistics.

In the tested benchmarks, the neighborhood operator subsumes spectral transform generators in class-separating power.

Conclusion

We investigated whether high-dimensional spectral representations are necessary for hyperspectral and multimodal land-cover classification once local spatial context is explicitly encoded. Across three benchmarks (Houston multimodal, Indian Pines, and PaviaU) using stratified 10% training splits, we found that a compact 10–14 dimensional neighborhood-statistics representation consistently outperforms 30–200 dimensional spectral baselines. Improvements range from +3.0 to +11.1 percentage points in overall accuracy and +6.3 to +16.1 points in average accuracy, with particularly strong gains for minority classes.

Performance saturates at a small, scene-dependent neighborhood scale (5×5–7×7), suggesting the existence of a

minimal sufficient spatial radius beyond which larger windows provide no consistent benefit. In contrast, uncertainty-driven multimodal gating strategies fail to improve performance and, in some cases, substantially degrade it, with gate confidence exhibiting slight negative correlation with classification correctness.

These results indicate that local second-order spatial statistics constitute a dominant and efficient representational mechanism for standard hyperspectral benchmarks. Importantly, this representation achieves superior performance with substantial dimensionality reduction (76–95% fewer features) and minimal computational overhead.

Future work should evaluate this principle on additional multimodal datasets, explore adaptive or anisotropic neighborhood selection, and analyze the interaction between explicit neighborhood statistics and learned convolutional representations.